

Case A: target-constrained execution

User objective

- task: *LD50 Quantitative toxicity regression*
- allowed methods: *PH, PL, CA*
- objective: satisfy $PCC^2 \geq 0.6$

Adaptive design and screening evidence

- LLM representation proposals: *hydrogen-inclusive 97.5% range vs. heavy-atom 90% range*
- Deterministic selection: *hydrogen-inclusive 97.5% range; 35 valid pair groups.*
- LLM subsampling proposal: *target-size stratified, $n \approx 750$; deterministic validation gives $n = 751$.*
- QC repair: *CA 0–18.16 Å $\rightarrow$ 0–9.64 Å; PH/PL reused unchanged.*
- Combination ranking: *PL(rank 1), PH+PL(rank 2).*

Progressive evaluation

- Candidate order is frozen from screening evidence.
- Evaluate rank-1 PL on full training / acceptance set.
- $PCC^2 = 0.607 \geq 0.6 \rightarrow$ target met $\rightarrow$ stop after the first full-scale candidate.

Output

- selected combination: *PL*
- acceptance score: $PCC^2 = 0.607$
- stop reason: *TARGET_MET*

TARGET MET

Case B: best-available execution

User objective

- task: *CASF-2016 protein-ligand affinity regression*
- allowed methods: *PH, PL, CA, FPRC, EIC*
- objective: *maximize over top 5 from screening; no target*

Adaptive design and screening evidence

- LLM representation proposals: *support-filtered 95% local range vs. conservative 90% local range.*
- Deterministic selection: *support-filtered 95% local range; pair support = 99.5%; 30 observed pair groups.*
- LLM subsampling proposal: *prefer a compact representative subsample with preserved target, size, and pair coverage.*
- Subsample sizes: *$n = 381, 750, 754$; smallest valid subsample $n = 381$ selected.*
- Combination ranking: *all 31 nonempty subsets are evaluated on the training subsample.*

Progressive evaluation

- Top-5 candidate order is frozen before acceptance evaluation.
- Evaluate all five candidates
- PCCs in screening order: *0.832, 0.826, 0.834, 0.832, 0.831.*
- Best: *CA + EIC + FPRC + PH (0.834); top-k exhausted $\rightarrow$ stop.*

Output

- selected combination: *(CA, EIC, FPRC, PH)*
- acceptance score: $PCC = 0.834$
- stop reason: *MAXIMIZATION_COMPLETE*

COMPLETE